9 **(a)** One assumption of the kinetic theory of gases is that the particles of the gas make perfectly elastic collisions among themselves and with the wall of the container. State two other assumptions of the kinetic theory of gases.

1. Any gas is made up of a large number of particles in constant and random motion.
2. The volume of each particle is negligible compared to the volume of the gas. This also means that any 2 particles are considered far apart (relative to their size).
3. The forces between the particles are negligible except during the time of collision. Having negligible forces between particles implies that the microscopic potential energy of the particles is constant and we can set it as 0 J.
4. The duration of collisions is negligible compared with the time interval between collisions.

(b) Consider a single molecule of mass *m* in a cuboidal container of internal side length L. The molecule is travelling with velocity *v* directly towards one of the walls W as shown in Fig 9.1.

Collisions between the molecule and the walls are elastic. This molecule makes multiple collisions in unit time. The pressure the molecule exerts on the wall is dependent on the frequency of the collisions with the wall.

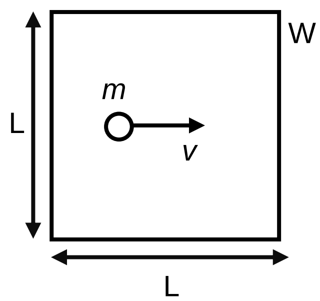

Fig. 9.1

Write expressions for

(i) the change in momentum of the molecule when it collides with wall W.

change in momentum $= p_f - p_i = -(mv) - (+mv) = -2mv$ .... [1]

(ii) time between collisions of the molecule and the wall W.

time between collisions $= \dfrac{\text{distance}}{\text{speed}} = \dfrac{2L}{v}$ .... [1]

(iii) the frequency of the collisions of the molecule with the wall W.

frequency of collisions $= \dfrac{1}{t} = \dfrac{v}{2L}$ .... [1]

(iv) the average momentum change per unit time for this molecule.

total momentum change = $f(\Delta p) = \dfrac{v}{2L}(-2mv) = -\dfrac{mv^2}{L}$ .... [1]

(v) the average force on wall W as a result of impacts by the molecule.

from (iv) and according to Newton's second law, the average force on the molecule $= \dfrac{mv^2}{L}$ [B1]

according to Newton's Third Law, average force on wall $= \dfrac{mv^2}{L}$ [B1]

[2]

(vi) the average pressure of wall W.

average pressure on wall $= \dfrac{\text{average force on wall}}{L^2} = \dfrac{mv^2}{L^3}$ [1]

(c) A cuboidal container contains *N* molecules of an ideal gas has volume V and pressure *p* given by the relation

$$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$$

where *m* is the mass of each molecule.

Explain briefly how the expression in **(b)(vi)** leads to this relation.

The volume of the container <u>*V* is equal to L^3</u>. [B1]

The equation was assuming every molecule was moving horizontally.

In a sample of gas of *N* molecules, each molecule could be moving randomly in any three directions within the container so that its actual velocity c is given by

$c^2 = c_x{}^2 + c_y{}^2 + c_z{}^2$

The molecules are moving with a <u>range of velocities,</u> $\langle c^2 \rangle$ is the mean square speed of the molecules, so $\langle c^2 \rangle = \langle c_x{}^2 \rangle + \langle c_y{}^2 \rangle + \langle c_z{}^2 \rangle$ [B1]

Since the number of particles is large and they are in <u>random motion</u>, the particles have an equal chance of being in any direction $\underline{\langle c_x^2 \rangle = \langle c_y^2 \rangle = \langle c_z^2 \rangle = \frac{1}{3}\langle c^2 \rangle}$ [B1]

The expression in **(b)(vi)** for average pressure due to one molecule moving in one direction is modified to be an expression for the average pressure due to <u>N molecules</u> in all directions.

average pressure on wall $= N\dfrac{m}{V}\left(\dfrac{1}{3}\langle c^2 \rangle\right) = \dfrac{1}{3}\dfrac{Nm\langle c^2 \rangle}{V}$ [B1] [4]

(d) **(i)** Use the equation of state for an ideal gas, together with the equation given in **(c)** to show that K, the average translational kinetic energy of a molecule is proportional to its absolute temperature [3]

$$pV = nRT$$

from (c) $$pV = \frac{1}{3}Nm\langle c^2\rangle$$

$$nRT = \frac{1}{3}Nm\langle c^2\rangle \quad \text{[M1]}$$

$$= \frac{1}{3}nN_A m\langle c^2\rangle$$

$$3RT = N_A m\langle c^2\rangle$$

$$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}\frac{R}{N_A}T \quad \text{[M1]}$$

$$pV = NkT$$

from (c) $$pV = \frac{1}{3}Nm\langle c^2\rangle$$

$$NkT = \frac{1}{3}Nm\langle c^2\rangle \quad \text{[M1]}$$

$$kT = \frac{1}{3}m\langle c^2\rangle$$

$$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT \quad \text{[M1]}$$

Since R and N_A are constants / Since k is constant [B1], the kinetic energy of a molecule is proportional to its absolute temperature. [A0]

(ii) Calculate , for oxygen and hydrogen at the same temperature, the ratio

$$\frac{\text{root mean square speed of a hydrogen molecule}}{\text{root mean square speed of an oxygen molecule}}.$$

The masses of the molecules are given the following table:

Mass of a hydrogen molecule = 3.34×10^{-27} kg

Mass of an oxygen molecule = 5.34×10^{-26} kg

Kinetic energy of a molecule is proportional to its absolute temperature.

So when temperature is the same, mean square speed is inversely proportional to mass

$$\frac{v_H^2}{v_O^2} = \frac{m_O}{m_H}$$

$$\frac{v_H}{v_O} = \sqrt{\frac{m_O}{m_H}} = \sqrt{\frac{5.34\times10^{-26}}{3.34\times10^{-27}}} \quad \text{[M1]}$$

$$= 3.98 \quad \text{[A1]}$$

ratio = = **…………………..** [2]

(iii) Suggest why the atmosphere of the Earth contains very little hydrogen.

Hydrogen molecules have higher speed [B1]
……………………………………………………………………………………………………

So a greater proportion/fraction of hydrogen molecules have a speed greater than
……………………………………………………………………………………………………

escape speed [B1]
……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

[Total: 20]

End of Paper